

# Augmenting the Human Perception of Comfort through Interactive AI

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## ABSTRACT

Comfort in work environments is highly influenced by indoor environmental quality—the combined effects of acoustics, thermal, lighting, air quality, and so forth. These physical parameters can impair productivity, and also can be a threat to health, and compromise the well-being. This dissertation work explores the opportunities for interactive artificial intelligence to bring radical changes in human experiences within built environments. The research description outlines four case studies that bring tailored notifications and actions to users in order to improve their comfort in the personal and social context in office buildings as well as at home.

## CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; *Empirical studies in interaction design*; Ubiquitous and mobile computing systems and tools.

## KEYWORDS

Indoor Environmental Quality, Human-Building Interaction, Interaction with Predictive Models

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## 1 CONTEXT AND MOTIVATION

Indoor Environmental Quality (IEQ) as a scientific research agenda seeks to reveal how various indoor conditions in a built environment affect its occupants' comfort, productivity, and health. Traditionally, these questions were subject to concerted investigations in the scientific domain of architecture to design guidelines and standards for building construction. However, in the current state of the art, it is increasingly recognized that these questions are essentially about human interaction [2]. Especially now that pandemic may cause a shift to work-from-home (WFH) in the future of work [23]. The vast past literature on teleworking focused on the tasks themselves (e.g., [12, 25, 46, 47]) and neglected the impact of the ambient environment, leaving space for further research to

separately address comfort in the individual and social context in the office building and at home.

This dissertation aims to bring interactive Artificial Intelligence (AI) [49] and Human-Computer Interaction (HCI) to the field of IEQ to push its boundaries, particularly by creating methods that can predict indoor environmental qualities. Most of the scientific attempts in this direction considered only one or two dimensions of human comfort (thermal, visual, etc.). This work considers inter-related dimensions of comfort to shed light on the quantitative and qualitative links between conditions around occupants and their comfort. Given the viewpoint that comfort is an interactive experience between humans and buildings, how can ambient intelligence raise users' awareness and change their behavior? How can the human agency reconcile building automation towards improving both comfort and sustainability measures? The role of computing in this context is to enable users to take timely actions. Answering these questions requires the training of intelligent personal agents, and designing an awareness system that can unobtrusively change long-term human experiences.

The motivation for this work is in two folds: *a)* Literature reflects that subjective ratings may not always serve as a standard to reflect the best IEQ for users themselves. On the one hand, the interpretation of the IEQ questionnaire could differ due to the variation of background and geographical location of the users [5, 38, 42], and subjective ratings collected in the experimental context and in the field settings may have discrepancies which cause consistency issues in the results [3]. On the other hand, for the IEQ dimension like air quality, human sensors may not be able to perceive the differences [31]. *b)* Increasing the intensity or operation time of mechanical building adjustment (e.g., ventilation) causes energy-efficiency issues, and many buildings are not equipped with automated solutions or do not provide a standard digital interface to the users, especially considering the users in developing regions in the field of HCI4D [1]. This work aims to make the users aware and engaged, prompting the change of routine and behavior also in the occasion where such sensing and predictive technology may not be available.

## 2 RELATED WORK

**HCI's approaches to Indoor Environmental Quality.** In recent years, interaction design research, and HCI more broadly, have offered diverse and remarkable contributions to address the complexity of human experiences with and within built environments (e.g., [13–15, 29, 36]). These contributions, unified as a growing branch of HCI—"Human-Building Interaction" (HBI) [2] which seeks to interface with the domains of Architecture and Urban Sciences to create a human-centered vision of "smart agenda" in built environments. The design of engaging interaction scenarios varies at least in four directions: *a)* "Ludic Design"—creating a

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medium for occupants to become curious, exploratory, and reflective with their environments and become thoughtful on how to make changes [16], *b*) "Optimally Informative"—making aware IEQ measurements through the most comprehensible and accessible means [10, 20, 21, 26, 30, 48, 52], *c*) "Gamified"—linking with social inclusiveness to strike a better connection between occupants' comfort and office activities [28], *d*) "General awareness"—pursuing a long-term impact on users via educational or peripheral communication [22, 44].

**Augmenting human sensors.** Augmenting people's senses with computational support can advance their ability to perceive information and perform tasks [35] yet how, when, and what to intervene are the key questions that HCI researchers try to answer when designing solutions to bridge physical space and information space [19], especially in ubiquitous computing [7]. Algorithmic and hardware-wise, the development of sensors to enable a smart environment is moving towards general-purpose and in-the-wild (e.g., [11, 24]); human and AI collaboration framework has been proposed to replace the traditional teacher-student framework (where human acts as a teacher and AI acts as a student) as the later one requires unrealistically abundant communication demands and teaching strategies from human trainer [4, 17]. Whereas concerning ethical issues [9], maintaining a reasonable level of human control gives mankind a chance to be acquainted with machine decisions [33] (e.g., [6]).

### 3 RESEARCH QUESTIONS

The overall goal of the dissertation work is to predict and prevent discomfort in the built environment using the smart technology. Specifically, the following research questions are addressed:

- (1) *RQ1*. What is the relationship between human subjective perception, human physiology, and environmental sensing in the context of comfort in the built environment?
- (2) *RQ2*. Can we predict discomfort with one or a combination of the three measurements in *RQ1*?
- (3) *RQ3*. How can human-machine dialog be designed to address the predicted discomfort and lead to behavior change?
- (4) *RQ4*. How can the complexity of the social context (e.g., doing individual work versus in a teleconference) be accommodated when we design the user interaction?

### 4 RESEARCH APPROACH AND METHODS

Long-term impact with lasting user sensitivity needs to be taken into consideration when creating design interventions for raising awareness or behavior change [43]. This is also the position of my dissertation work—developing intelligent systems to complement the individual's perceptual ability [50, 51], augmenting user's awareness and sensitivity towards the quality of indoor environments they inhabit in their daily routine, and supporting human reasoning styles [8] with respect to the evolution of modern workspace. Specifically, following the human-centered artificial intelligence framework [41], this dissertation work articulates two positions of interactive AI in the built environment: *a*) "*Artificial sensing*": For the IEQ parameters that users could not sense instantly but can cause discomfort, the system will monitor, forecast, and inform users of the situation through appropriate means. *b*) "*Augmented*

*human sensing*": For IEQ parameters that users can – to some extent – sense but cannot subjectively reflect the differences immediately, the design intervention follows the exploitation and exploration learning process [27]. It aims to observe and learn from users' daily routine to understand users' preferences and attitudes about the built environment (i.e. exploitation) and at the same time measure users' physiological reactions towards the environmental condition and propose suggestions that may improve the comfort situation (i.e. exploration).

To examine the performance of the above-described system on acknowledging the IEQ conditions to the users yet retaining human agency in the control, I am on the track of conducting four case studies to address "artificial sensing" and "augmented human sensing" interventions in personal and social contexts separately (see Figure 1), among which case study I and II have been accomplished, case study III and case IV are ongoing work till the end of my Ph.D. period. The remainder of this section illustrates the implementation details of the two positions. The four case studies share the same aim to answer *RQ2* to *RQ3*. Within them, case study II and IV address *RQ4*. At the same time, the data collected during our experiments and surveys provide answers to *RQ1*.

**"Artificial sensing".** This is a multi-stage process centered around users' preferences towards possible interactions. Indoor air quality (specifically, carbon dioxide concentration level, which is commonly used as an indicator for assessing ventilation efficiency and consequently the overall Indoor Air Quality (IAQ) [18, 34, 37, 45]) was chosen as the studied environmental parameter since it is an indoor parameter that users have difficulties reflecting on it. In the first stage, field data collection was accomplished in two office buildings that host more than 300 employees; 24 environmental sensing devices were deployed at 28 individual desks and 18 meeting rooms in two periods between 14th October 2019 and 13th March 2020 in Switzerland. In the second stage, prototypes were created, adhered to the iterative design process where descriptive data analysis was performed and preliminary predictive models were developed based on the collected data. It was followed by controlled user studies and surveys to understand users' considerations towards proposed design solutions in the different social and environmental contexts. For the personal context, three visualizations were designed to communicate the risk, urgency, and uncertainty of predicted IAQ presented via Apple watch (see case study I in Figure 1). For the social context, a design solution was proposed to communicate CO<sub>2</sub> risks right at the beginning of the meeting and during the meeting (see case study II in Figure 1). More details will be discussed in Section 5.

**"Augmented human sensing".** In this phase, the studied environmental parameter has been shifted to visual comfort since it is a parameter that human sensors can capture but have difficulty reflecting on. A study found that even if users can notice the change of lighting condition, they seem to have difficulty on when and what to act on if they are concentrating on work [32]. To date, I have conducted a literature review on visual comfort measures and run an online survey on 479 office workers. The ongoing analysis indicates that there is interdependence between psychological (e.g., mental state caused by task difficulty), physiological (e.g., eye strain), social (e.g., whether the user is situated in a collaborative condition like

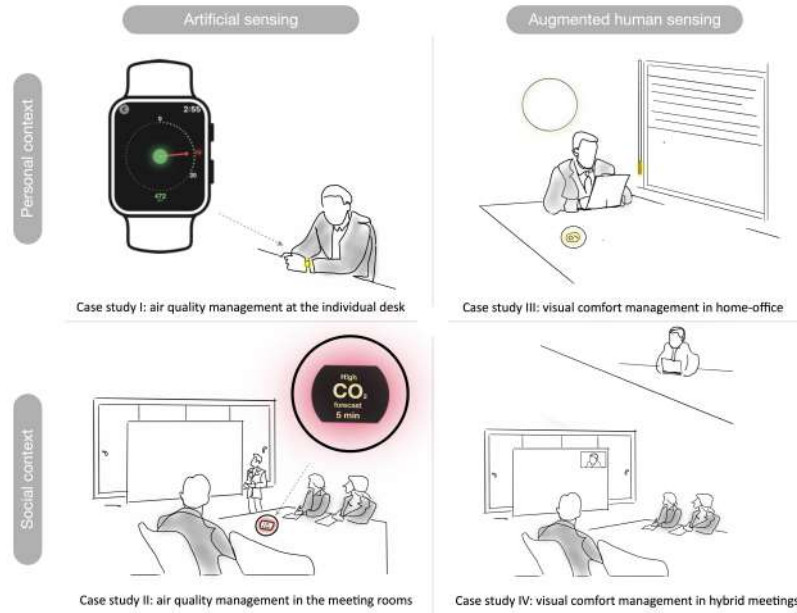


Figure 1: Planned case studies to examine how interactive AI can assist users to manage indoor comfort.

teleconferencing), and technological factors (e.g., screen resolution) concerning visual comfort. Preliminary results reveal the need to advance a comfort-aware home-office environment and be prepared for hybrid meeting scenarios. Building upon the survey and literature review, the expected next steps, namely the execution of the case study III, will start with investigating wearable and ubiquitous methods for detecting visual fatigue which is reflected by users' physiological reactions (e.g., blink rate [39], eyelid squinting [40], gaze, pupil diameter signal, saccade and fixation, and body movements) followed by the design of a notification system to assist the users to manage their visual comfort (see Figure 2 for example propositions). The last case study is planned to investigate visual comfort in the social context, specifically, during hybrid meetings.

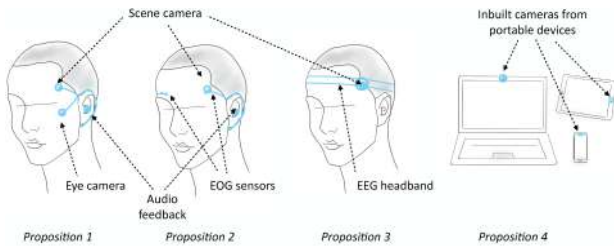


Figure 2: Possible sensory artifacts and software: *Proposition 1-3* are wearable interventions and *Proposition 4* is a ubiquitous intervention.

## 5 RESULTS TO DATE

An indoor environmental sensing device, co-developed by our industry partner, was designed to measure temperature, humidity, air pressure, CO<sub>2</sub>, particulate matters, airflow, illuminance, and ambient noise level. During the data collection phase, 763 subjective

ratings were collected and the current data revealed that there is no direct correlation between the change of environmental conditions and perception of comfort<sup>1</sup>. Aligned with the literature, users had difficulty sensing air quality changes as it is invisible and odorless. However, due to the experiment period (late autumn to early spring), the data collection for temperature in relation to subjective rating could be incomplete, and parameters that impact visual comfort like flicker and glare [3] were not captured in this phase which leaves space for further research.

Case study I has been performed by designing three interfaces, grounded in the concepts of *Risk*, *Temporal Proximity*, and *Certainty* (see Figure 3) [53]. The conducted user study revealed that the visualization of indoor CO<sub>2</sub> forecasts could support them in taking preventative actions. The results also shed light on necessary future improvements, such as incorporating information that can help users strike a balance between air quality, thermal comfort, and energy use.

Case study II has been carried out by studying 1007 meeting sessions [54]. The descriptive analysis revealed that the occupants spent 66% of the meeting time in situations where the level of CO<sub>2</sub> passes the threshold above which adverse impact on well-being and cognitive capacities can be expected. Capitalizing on the collected data, seven patterns of CO<sub>2</sub> evolution during the meetings were elicited (see Figure 4b). With the prior knowledge of how the CO<sub>2</sub> will evolve during the meeting, interactive solutions that would minimize the cost of notification and interruption of the meeting were designed, informed by two online surveys that examine the impact of timing and modality of interaction on users' decision (see Figure 4a).

<sup>1</sup> Spearman's correlation of CO<sub>2</sub> (ranging from 278 to 1915 ppm) and subjective rating (-4 intolerable to 0 comfortable): -0.162; illuminance (8 to 7197 lux): 0.008; temperature (22 to 34 °C): -0.123; noise (19 to 61 dB(A)): -0.024.

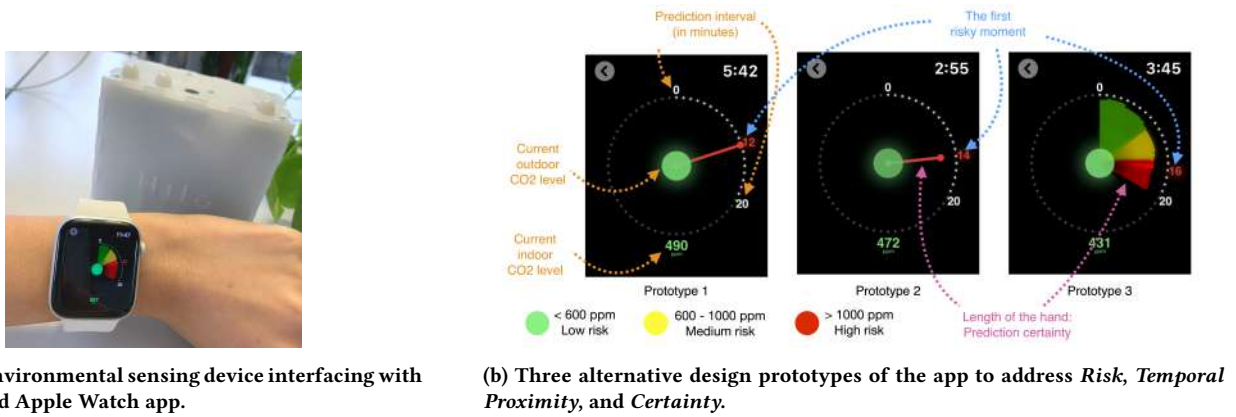


Figure 3: Prototypes in case study I.

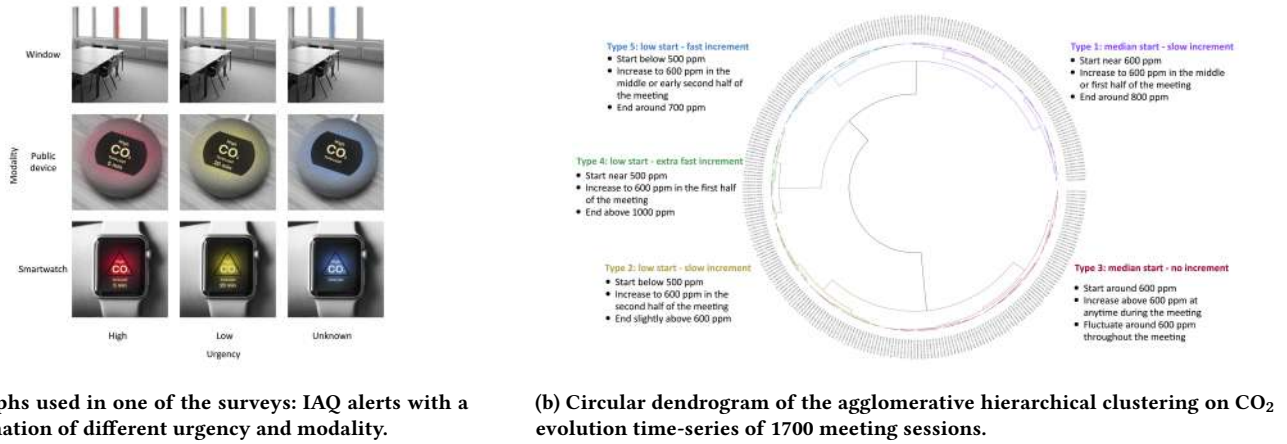


Figure 4: Results from case study II.

## 6 CURRENT AND EXPECTED CONTRIBUTIONS

Through this dissertation work, I expect to make the following contributions:

- (1) Field data on human subjective perception, human physiology, and environmental sensing.
- (2) New predictive algorithms or frameworks for discomfort forecast in the built environment.
- (3) Design instances/artifacts for comfort dialog in the built environment.
- (4) Design implications of IEQ communication in four social contexts: at individual desks and in the meetings; in present and during teleworking.

The dissertation work argues that Human-Building Interaction is a personal experience, and given the current trend of automation, it is necessary to broaden the consideration of raising users' awareness as well as offering user control over the "intelligent and interactive" environments. Hence, the dissertation work also holds an additional broader contribution to the relations and interdependencies between human sensing and machine sensing, examining

intelligently-augmented human awareness as an interaction design research challenge.

## 7 DISSERTATION STATUS AND LONG TERM GOALS

I am currently in the third year of the Ph.D. program, which holds an expected completion date of mid-to-end 2023. In the first year, a general literature review of comfort studies in the built environment, and focused research on indoor air quality were conducted. Data collection by the first prototype of the indoor environmental sensing device has been organized in two buildings. Two case studies have been accomplished which resulted in one late-breaking work and one full paper. In the second year, more review has been done on visual comfort and in the personal and social contexts. In the third and fourth years, design proposals for visual comfort control in these two contexts are planned to be developed and experimented with. In the long run, I plan to continue my research in comfort management to further explore the interaction of various dimensions of comfort (e.g., thermal and acoustic). Furthermore, I hope to extend my work to vulnerable groups (e.g., people with

special needs and disabilities) to increase accessibility and inclusion in the interaction with built environments.

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